

TCP vs UDP — Detailed Interview Notes

Introduction

Technical Definition

Transmission Control Protocol

TCP (Transmission Control Protocol) is a connection-oriented transport layer protocol that provides reliable, ordered, and error-checked data delivery between devices.

Technical Definition

User Datagram Protocol

UDP (User Datagram Protocol) is a connectionless transport layer protocol that sends data without establishing a connection and without guaranteeing delivery.

Why Transport Layer Protocols are Needed?

When data travels over network:

- Data may get lost
- Data may arrive out of order
- Data may get corrupted

Transport layer protocols solve these problems.

Main protocols:

- TCP
- UDP

Both work at:

- Transport Layer (OSI Layer 4)
-

TCP (Transmission Control Protocol)

TCP focuses on:

- Reliability
- Accuracy
- Ordered delivery

TCP ensures data reaches destination correctly.

Important Features of TCP

1. Connection-Oriented

Before sending data:

- Connection established
- Communication starts
- Connection terminated afterward

This process is called:

TCP Three-Way Handshake.

TCP Three-Way Handshake

Used to establish reliable connection.

Steps:

Step 1 — SYN

Client sends:

"SYN"

Meaning:

"I want to establish connection."

Step 2 — SYN-ACK

Server replies:

"SYN + ACK"

Meaning:

"Connection request accepted."

Step 3 — ACK

Client sends:

"ACK"

Connection established.

Visualization

Client → SYN

Server → SYN ACK

Client → ACK

Connection starts.

Why Handshake Needed?

Ensures:

- Both devices ready
 - Reliable communication possible
-

2. Reliable Communication

TCP guarantees delivery.

If packet lost:

- TCP retransmits packet

This makes TCP reliable.

3. Ordered Delivery

Packets may travel through different paths.

TCP rearranges them correctly.

Example:

Message:

HELLO

Even if packets arrive:

- HLOEL

TCP rearranges:

HELLO

4. Error Checking

TCP checks:

- Missing packets
- Corrupted packets

Uses:

Checksum mechanism.

5. Flow Control

Prevents sender from overwhelming receiver.

Uses:

Sliding Window Protocol.

6. Congestion Control

Controls network traffic during congestion.

Avoids network overload.

Applications Using TCP

Because TCP is reliable, it is used where accuracy is important.

Examples:

- Web browsing
- Email
- File transfer
- Banking systems

Protocols using TCP:

- HTTP
 - HTTPS
 - FTP
 - SMTP
-

UDP (User Datagram Protocol)

UDP focuses on:

- Speed
- Low delay

UDP does NOT guarantee delivery.

Features of UDP

1. Connectionless

No connection establishment.

Packets sent directly.

No handshake.

2. Faster Communication

No reliability checking.

No acknowledgments.

Therefore:

Very fast.

3. No Ordered Delivery

Packets may arrive:

- Out of order
- Lost
- Duplicated

UDP does not fix these problems.

4. No Error Recovery

If packet lost:

UDP ignores it.

No retransmission.

Applications Using UDP

UDP used where:

- Speed more important than accuracy

Examples:

- Video streaming
 - Online gaming
 - Live broadcasting
 - Voice calls
-

Why UDP Used in Video Calls?

Suppose one packet lost in video call.

Better to skip packet than delay whole video.

Therefore:

UDP preferred.

TCP vs UDP — Major Differences

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Speed	Slower	Faster
Error Checking	Yes	Minimal
Ordered Delivery	Yes	No
Acknowledgment	Yes	No
Flow Control	Yes	No
Congestion Control	Yes	No

Real Life Analogy

TCP Analogy

Courier service requiring signature.

Ensures package delivered correctly.

UDP Analogy

Radio broadcasting.

No confirmation whether listener received message.

Important Interview Questions

Difference between TCP and UDP?

TCP is reliable and connection-oriented.

UDP is fast and connectionless.

Why TCP slower?

Because it performs:

- Handshake
 - Error checking
 - Acknowledgment
 - Retransmission
-

Why UDP faster?

No connection setup or reliability mechanisms.

Which applications use UDP?

Gaming, streaming, VoIP.

Which applications use TCP?

Web browsing, email, file transfer.

OSI vs TCP/IP — Detailed Interview Notes

Introduction to Network Models

Network communication is complex.

Models help standardize communication.

Two important models:

- OSI Model
 - TCP/IP Model
-

OSI Model

Technical Definition

OSI Model

OSI (Open Systems Interconnection) model is a conceptual framework developed to standardize networking communication into seven layers.

Developed by:

- ISO organization

Purpose of OSI Model

Helps:

- Understand networking
- Standardize communication
- Troubleshoot networks

7 Layers of OSI Model

1. Physical Layer

Responsible for:

- Transmission of bits

Devices:

- Hub
- Repeater

2. Data Link Layer

Responsible for:

- Framing
- MAC addressing
- Error detection

Devices:

- Switch
- Bridge

3. Network Layer

Responsible for:

- Routing
- IP addressing

Device:

- Router

Protocols:

- IP
-

4. Transport Layer

Responsible for:

- Reliable delivery
- Segmentation

Protocols:

- TCP
 - UDP
-

5. Session Layer

Responsible for:

- Managing sessions
-

6. Presentation Layer

Responsible for:

- Encryption
 - Compression
 - Data translation
-

7. Application Layer

Provides user services.

Protocols:

- HTTP
 - FTP
 - SMTP
-

TCP/IP Model

Technical Definition

TCP/IP

TCP/IP model is the practical networking model used on the Internet.

Developed by:

- DARPA
-

Layers of TCP/IP

1. Network Access Layer

Combines:

- Physical
 - Data Link
-

2. Internet Layer

Equivalent to:

- Network Layer

Protocol:

- IP
-

3. Transport Layer

Protocols:

- TCP
 - UDP
-

4. Application Layer

Combines:

- Session
 - Presentation
 - Application
-

OSI vs TCP/IP Differences

Feature	OSI	TCP/IP
Layers	7	4
Type	Conceptual	Practical
Developed By	ISO	DARPA
Usage	Reference model Internet communication	
Session Layer	Separate	Combined
Presentation Layer	Separate	Combined

Why TCP/IP More Popular?

Because:

- Used on Internet
- Simpler
- Practical implementation exists

OSI mostly used for:

- Learning
- Troubleshooting

Important Interview Questions

Difference between OSI and TCP/IP?

OSI has 7 layers and is conceptual.
TCP/IP has 4 layers and is practical.

Which model used on Internet?

TCP/IP.

Which layer performs routing?

Network Layer.

Which layer uses TCP and UDP?

Transport Layer.

HTTP vs HTTPS — Detailed Interview Notes

HTTP

Technical Definition

Hypertext Transfer Protocol

HTTP (Hypertext Transfer Protocol) is an application layer protocol used for communication between web browsers and web servers.

Purpose of HTTP

HTTP transfers:

- Web pages
- Images
- Videos
- Data

between:

- Client (browser)
 - Server
-

How HTTP Works?

1. User enters URL
 2. Browser sends HTTP request
 3. Server processes request
 4. Server sends HTTP response
 5. Browser displays webpage
-

Problem with HTTP

HTTP sends data:

- In plain text

Meaning:

Anyone intercepting network can read data.

Dangerous for:

- Passwords

- Banking
 - Personal information
-

HTTPS

Technical Definition

HTTPS

HTTPS (Hypertext Transfer Protocol Secure) is a secure version of HTTP that uses SSL/TLS encryption for secure communication.

SSL/TLS

Technical Definition

Transport Layer Security

TLS/SSL provides:

- Encryption
 - Authentication
 - Secure communication
-

Why HTTPS is Secure?

HTTPS encrypts data before transmission.

Even if attacker captures packets:
Cannot read actual data.

HTTPS Security Features

1. Encryption

Data converted into unreadable form.

2. Authentication

Ensures website is genuine.

3. Data Integrity

Prevents modification during transmission.

HTTPS Working

1. Browser requests secure connection
 2. SSL/TLS handshake occurs
 3. Encryption keys exchanged
 4. Secure communication begins
-

HTTP vs HTTPS Differences

Feature	HTTP	HTTPS
Security	No	Yes
Encryption	No	Yes
Port	80	443
Certificate	Not required	SSL certificate required
Speed	Slightly faster	Slightly slower
Usage	Normal sites	Secure websites

Real Life Analogy

HTTP

Sending postcard.

Anyone can read message.

HTTPS

Sending locked envelope.

Only receiver can read message.

Important Interview Questions

Difference between HTTP and HTTPS?

HTTPS provides encrypted secure communication using SSL/TLS.

Why HTTPS important?

Protects sensitive data.

What is SSL/TLS?

Security protocols used for encryption.

Which port HTTPS uses?

443.

Which layer HTTP works on?

Application Layer.

DNS (Domain Name System) — Detailed Notes with Diagram

Introduction to DNS

Technical Definition

Domain Name System

DNS (Domain Name System) is a distributed hierarchical naming system that translates human-readable domain names into IP addresses so computers can communicate over networks.

Why DNS is Needed?

Humans remember names easily.

Examples:

- google.com
- youtube.com
- openai.com

But computers understand only:

- IP addresses

Example:

142.250.183.78

DNS acts as translator between:

- Human language
 - Computer language
-

Real Life Analogy

DNS works like:
Mobile contact list.

You search:
"Papa"

Phone internally finds:
Actual phone number.

Similarly:
Browser searches:
google.com

DNS finds:
Corresponding IP address.

Problem Without DNS

Without DNS:
Users must remember IP addresses of every website.

Example:
Instead of typing:
google.com

You would type:
142.250.183.78

This would be very difficult.

Basic DNS Working

When user types:
google.com

DNS converts it into:
IP address

Then browser connects to actual server.

DNS Resolution Process (Very Important)

Step-by-Step Working

Step 1 — User Enters Domain Name

User types:
google.com

in browser.

Step 2 — Browser Cache Check

Browser first checks:

“Do I already know this IP?”

If already stored:

Connection happens immediately.

If not:

Next step starts.

Step 3 — Operating System Cache

OS checks local DNS cache.

If not found:

Request moves forward.

Step 4 — DNS Resolver

Request goes to:

DNS Resolver

Usually provided by:

- ISP
 - Google DNS
 - Cloudflare DNS
-

Step 5 — Root DNS Server

Resolver contacts:

Root DNS Server.

Root server says:

“I don’t know exact IP, but ask .com server.”

Step 6 — TLD Server

TLD = Top Level Domain server.

For:

google.com

TLD server is:

.com server

TLD says:

“Ask authoritative server.”

Step 7 — Authoritative DNS Server

Authoritative server stores actual IP.

It returns:

142.250.xxx.xxx

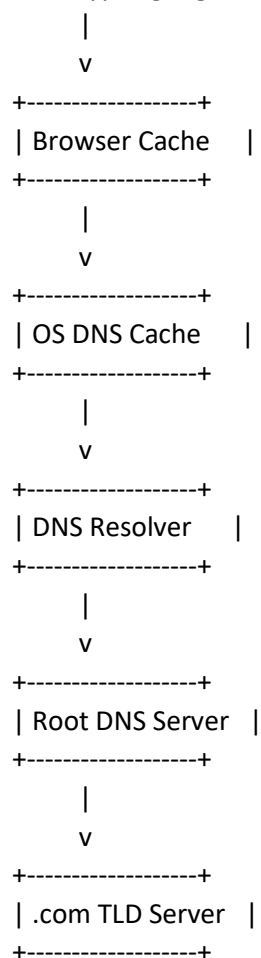
Step 8 — Browser Connects

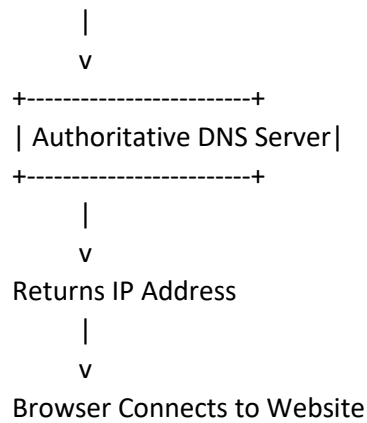
Browser receives IP address and connects to website server.

Webpage loads.

Full DNS Flow Diagram

User Types google.com





Types of DNS Servers

1. Root DNS Server

Top-level server in DNS hierarchy.

Knows:

Where TLD servers are located.

2. TLD Server

TLD = Top Level Domain

Examples:

- .com
- .org
- .in

TLD server knows:

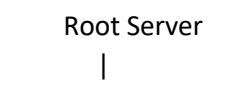
Authoritative server location.

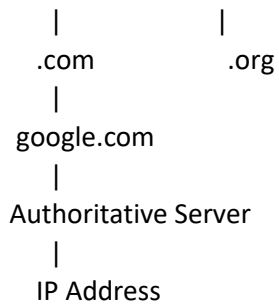
3. Authoritative DNS Server

Stores actual domain-IP mapping.

This server provides final answer.

DNS Hierarchy Diagram





Important DNS Records

DNS stores records.

1. A Record

Maps:

Domain → IPv4 address

Example:

google.com → 142.250.183.78

2. AAAA Record

Maps:

Domain → IPv6 address

3. CNAME Record

Alias for another domain.

Example:

mail.google.com → google.com

4. MX Record

Mail exchange record.

Used for:

Email servers.

5. NS Record

Specifies authoritative name server.

DNS Caching

DNS stores previously resolved addresses temporarily.

Benefits:

- Faster access
 - Reduced network traffic
 - Reduced DNS load
-

DNS Cache Example

First request:

May take longer.

Second request:

Faster because IP already cached.

Recursive vs Iterative Query

Recursive Query

DNS resolver finds complete answer for client.

Client gets final IP only.

Iterative Query

DNS server gives partial answer and redirects further.

DNS Uses UDP or TCP?

Usually:

DNS uses UDP Port 53

Because:

- Faster
- Lightweight

Sometimes:

TCP used for:

- Large responses
- Zone transfer

DNS Security Issues

DNS can be attacked.

Examples:

- DNS Spoofing
- DNS Poisoning
- DDoS attacks

DNS Spoofing

Attacker sends fake DNS response.

User redirected to fake website.

DNSSEC

Technical Definition

DNS Security Extensions

DNSSEC adds authentication and integrity protection to DNS responses.

Helps prevent:

- Spoofing
- Tampering

Advantages of DNS

- Easy website access
- Human-friendly names
- Faster browsing with caching
- Distributed architecture

Disadvantages of DNS

- DNS attacks possible
 - Initial lookup delay
 - Complex infrastructure
-

Important Ports

Service	Port
----------------	-------------

DNS UDP	53
---------	----

DNS TCP	53
---------	----

Real Life Example

When opening:
youtube.com

DNS:

1. Finds IP
2. Browser connects to YouTube server
3. Video loads

Without DNS:

Internet browsing would become extremely difficult.

Most Important Interview Questions

What is DNS?

DNS translates domain names into IP addresses.

Why DNS needed?

Humans remember names easier than IP addresses.

Which port DNS uses?

Port 53.

DNS uses TCP or UDP?

Mostly UDP.

What is DNS cache?

Temporary storage of resolved addresses.

What is Authoritative DNS Server?

Server containing actual domain-IP mapping.

What is DNS Spoofing?

Attacker sends fake DNS response.

Difference between recursive and iterative query?

Recursive	Iterative
Full answer returned	Partial answer returned

What happens when you type google.com?

1. Browser checks cache
2. DNS lookup occurs
3. IP address returned
4. Browser connects to server
5. Website displayed

Routing — Detailed Notes with Diagram

Introduction to Routing

Technical Definition

Routing

Routing is the process of selecting the best path for transferring data packets from source to destination across multiple networks.

Why Routing is Needed?

Suppose:

- Your computer in India sends request to Google server in USA.

There are:

- Thousands of routers
- Multiple paths
- Different networks

Question:

How does packet know where to go?

Answer:

Routing.

Routers decide:

- Best path
- Fastest path
- Available path

for packet delivery.

Real Life Analogy

Routing is like:

Google Maps navigation.

When going somewhere:

- Many roads exist
- Maps selects best route

Similarly:

Router selects best network path.

What is a Router?

Technical Definition

Router

Router is a Layer 3 networking device that forwards packets between different networks using IP addresses.

Router works on:

- Network Layer
-

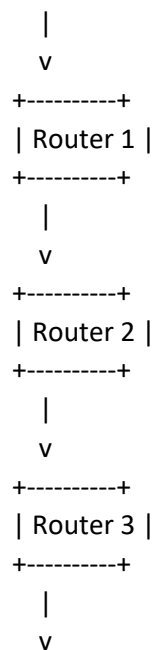
Basic Working of Routing

When router receives packet:

1. Reads destination IP address
2. Checks routing table
3. Finds best path
4. Forwards packet

Routing Process Diagram

Computer A



Destination Server

Packets travel through multiple routers.

Routing Table

Router stores routes inside:
Routing Table.

Routing Table Example

Destination Network Next Hop

192.168.1.0	Port 1
10.0.0.0	Port 2

Router uses this table to forward packets.

Types of Routing

1. Static Routing

Technical Definition

Static Routing

Static routing is routing where paths are manually configured by network administrator.

Features

- Manual configuration
 - Fixed path
 - Simple networks
-

Advantages

- Secure
 - Less CPU usage
 - Predictable
-

Disadvantages

- Difficult management
 - Not scalable
 - No automatic updates
-

Example

Admin manually sets:

“To reach Network B use Router X.”

2. Dynamic Routing

Technical Definition

Dynamic Routing

Dynamic routing automatically updates routes using routing protocols.

Why Dynamic Routing Needed?

Large networks change frequently.

Dynamic routing:

- Detects failures

- Finds alternate paths
 - Updates automatically
-

Advantages

- Automatic updates
 - Better scalability
 - Fault tolerance
-

Disadvantages

- More complexity
 - Higher resource usage
-

Dynamic Routing Protocols

1. RIP

Routing Information Protocol.

Uses:

Hop count.

Maximum:

15 hops.

2. OSPF

Open Shortest Path First.

Uses:

Shortest path algorithm.

Very fast and efficient.

3. BGP

Border Gateway Protocol.

Used on:

Internet backbone.

Controls routing between large networks.

Metrics in Routing

Routers use metrics to choose best route.

Examples:

- Hop count
- Delay
- Bandwidth
- Cost

Hop Count

Number of routers packet crosses.

Lower hop count usually preferred.

Default Gateway

Technical Definition

Default Gateway

Default gateway is the router used to send packets outside local network.

Example

Home WiFi router acts as:

Default gateway.

Packet Forwarding Process

Example Scenario

Computer A wants to send data to Google.

Process:

1. Checks destination IP
2. Sends packet to router
3. Router checks routing table
4. Packet forwarded through Internet routers

5. Reaches Google server

Routing Algorithms

1. Distance Vector Algorithm

Routers share routing information with neighbors.

Example:

RIP.

2. Link State Algorithm

Router builds complete network map.

Example:

OSPF.

Difference Between Distance Vector & Link State

Distance Vector Link State

Simple	Complex
Slower	Faster
Less scalable	Highly scalable

Routing Problems

1. Routing Loop

Packet keeps circulating endlessly.

2. Congestion

Too much traffic causes delays.

3. Packet Loss

Packets dropped due to overload.

Important Interview Questions

What is routing?

Process of selecting best path for packets.

Which device performs routing?

Router.

Which layer routing works on?

Network Layer.

Difference between static and dynamic routing?

Static	Dynamic
Manual	Automatic
Small networks	Large networks

What is default gateway?

Router used to access external networks.

Which protocol used on Internet?

BGP.

What is hop count?

Number of routers crossed.

Subnetting — Detailed Notes with Diagram

Introduction to Subnetting

Technical Definition

Subnetting

Subnetting is the process of dividing a large network into smaller logical subnetworks called subnets.

Why Subnetting is Needed?

Suppose one company has:
5000 computers.

If all devices placed in one network:

- Traffic becomes huge
- Broadcasts increase
- Performance decreases
- Security problems occur

Subnetting solves these problems.

Benefits of Subnetting

- Reduces traffic
 - Improves performance
 - Better security
 - Easier management
 - Efficient IP usage
-

Real Life Analogy

Imagine:
One giant classroom with 500 students.

Very noisy.

Solution:
Divide into smaller classes.

Subnetting works similarly.

IP Address Review

IPv4 Structure

IPv4 address:
32 bits.

Example:
192.168.1.10

Split into:

- Network part

- Host part

Binary Representation

192.168.1.10

11000000.10101000.00000001.00001010

Network ID and Host ID

Example:

192.168.1.10/24

- Network ID → 192.168.1
 - Host ID → 10
-

Subnet Mask

Technical Definition

Subnet Mask

Subnet mask determines which part of IP address belongs to network and which part belongs to host.

Example

IP:

192.168.1.10

Subnet Mask:

255.255.255.0

Meaning:

First 24 bits = Network

Last 8 bits = Host

CIDR Notation

CIDR:

Classless Inter-Domain Routing.

Example:

192.168.1.0/24

/24 means:

24 bits reserved for network.

Common CIDR Values

CIDR Subnet Mask

/8 255.0.0.0

/16 255.255.0.0

/24 255.255.255.0

Subnetting Example

Suppose:

192.168.1.0/24

Need:

4 subnets.

Borrowing Bits

Original:

24 network bits.

Need 4 subnets:

Borrow 2 bits.

New subnet:

192.168.1.0/26

Why /26?

$2^2 = 4$ subnets.

New Subnets

Subnet	Range
192.168.1.0/26	0–63
192.168.1.64/26	64–127
192.168.1.128/26	128–191
192.168.1.192/26	192–255

Subnet Diagram

Main Network
192.168.1.0/24

|

| | | |

Subnet1 Subnet2 Subnet3 Subnet4

Important Terms

1. Network Address

First address of subnet.

Used to identify subnet.

2. Broadcast Address

Last address of subnet.

Used to send data to all devices.

3. Usable Host Range

Addresses between:

- Network address
- Broadcast address

Example

Subnet:
192.168.1.0/26

Type	Address
Network Address	192.168.1.0
First Host	192.168.1.1
Last Host	192.168.1.62
Broadcast Address	192.168.1.63

Formula for Subnetting

Number of Subnets

$$2^n$$

Where:

n = borrowed bits

Number of Hosts

$$2^h - 2$$

Where:

h = host bits

Subtract 2 because:

- Network address reserved
 - Broadcast address reserved
-

VLSM (Variable Length Subnet Mask)

Technical Definition

Variable Length Subnet Mask

VLSM allows creation of subnets with different sizes based on requirements.

Why VLSM Needed?

Different departments need different hosts.

Example:

- HR → 20 hosts
- IT → 200 hosts

VLSM saves IP addresses.

FLSM vs VLSM

FLSM

VLSM

Equal subnet sizes

Variable sizes

Less efficient

More efficient

Advantages of Subnetting

- Better performance
- Reduced broadcast traffic
- Better security
- Efficient IP usage

Disadvantages

- Complex calculations
- Requires planning

Important Interview Questions

What is subnetting?

Dividing large network into smaller networks.

Why subnetting needed?

Reduce traffic and improve performance.

What is subnet mask?

Separates network and host portion.

What is CIDR?

Classless Inter-Domain Routing notation.

Difference between network and broadcast address?

Network Address	Broadcast Address
Identifies subnet	Sends to all hosts

Formula for number of hosts?

$2^h - 2$